

1 Data

PhysioNet EEGMMIDB

109 subjects, 64 ch, 160 Hz

imagery runs 4, 8, 12

epochs 0.5–2.5 s

rest, left, right

2 Splits

30 targets, 6 groups of 5

pool per group: 98 others

target epochs in 3

rotated partitions:

2 calibrate, 1 test

3 Nested montages

$64 \supset 32 \supset 16 \supset 8 \supset 4 \supset 2$

by distance to C3/C4

(8 adds Cz and Fz)

nesting checked;

the run aborts if broken

4 One pipeline

CAR, 4–40 Hz, μV ,

Tukey epoch rejection

8 decoders trained

on the pool, calibrated

on 48 target epochs

5 Two levels

3-class output read as

L1 detection: rest vs MI,

balanced accuracy

L2 lateralization: L vs R,

above 0.5 ($1 - \text{FNR}$)

Paired Wilcoxon signed-rank tests with the subject as the unit ($n = 30$), two-sided $\alpha = 0.05$; one seed (42).

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